



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Preliminary Analysis of B2B Courier Charge Accuracy Using ML

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Guide Details

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Mini Project Outlines

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Introduction

- Rapid growth of e-commerce and supply chain systems has increased demand for efficient logistics.
- In B2B logistics, courier services are essential for transporting goods between businesses.
- Accurate estimation of courier charges is critical for:
 - Cost control
 - Billing transparency
 - Building trust between clients and service providers
- These inconsistencies lead to:
 - Financial losses
 - Operational inefficiencies
 - Reduced customer satisfaction

Existing System Vs Proposed

Existing System

- Uses predefined pricing rules and rate cards.
- Relies on manual billing and verification.
- Uses basic statistical methods (like linear regression).
- Limited use of historical data.
- Leads to billing errors and low accuracy.

Proposed System

- Uses machine learning algorithms (e.g., Random Forest).
- Based on historical data analysis.
- Provides automated prediction and validation.
- Handles complex and non-linear relationships.
- Improves accuracy and reduces billing errors.

Research Gaps

- Focus mainly on cost prediction, not validation of actual vs estimated charges.
- Limited research on B2B courier billing accuracy.
- Traditional models fail to capture complex & non-linear patterns.
- Lack of real-time discrepancy detection systems.
- Limited use of advanced ML models (e.g., Random Forest).
- No focus on identifying causes of charge deviations.
- Few systems integrate prediction + validation + analysis.



Problem Statement

- Accurate courier charge estimation is critical in **B2B logistics**.
- Existing systems rely on **predefined rules and basic models**
- Unable to handle **dynamic factors** (weight, distance, service type, pricing policies).
- Leads to **discrepancies between estimated and actual charges**.



Proposed Project Objectives

- To analyze historical B2B courier data to understand pricing patterns
- To identify deviations between estimated and actual courier charges
- To study key influencing factors such as weight, distance, service type, and carrier
- To develop a machine learning model for accurate courier charge prediction
- To compare and evaluate different ML algorithms for better performance

Proposed Project Outcomes

- To implement an automated system for charge validation
- To visualize results using charts and dashboards for better understanding
- To provide insights for reducing billing errors and improving transparency

Software and Hardware Requirements

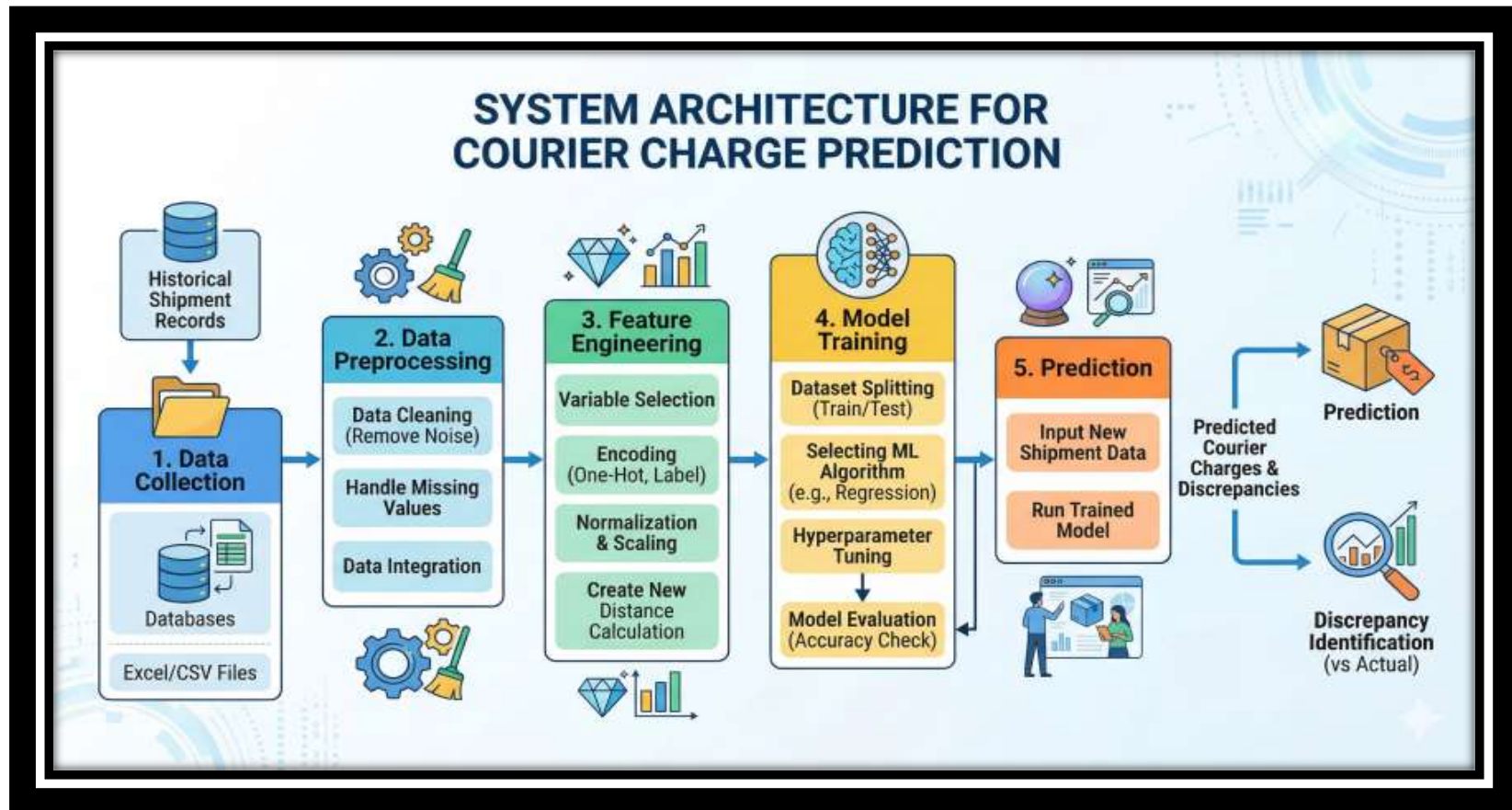
Hardware Requirements

- Processor: Intel i5 or above
- RAM: Minimum 8 GB (Recommended: 16 GB)
- Storage: Minimum 256 GB HDD/SSD
- System Type: 64-bit system

Software Requirements

- Operating System: Windows / Linux / macOS
- Programming Language: Python
- Libraries: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn
- Development Environment: Jupyter Notebook / Google Colab / VS Code
- Database (Optional): MySQL / SQLite

Architectural Diagram



System Architecture for Machine Learning-Based Courier Charge Prediction and Discrepancy Detection

PROPOSED METHOD AND ALGORITHMS

- **Proposed Method**
- Collect courier dataset
- Preprocess data (clean & encode)
- Select important features
- Train ML model (Random Forest)
- Predict and validate courier charges

- **Algorithm**
- Load dataset
- Preprocess data
- Split into train & test
- Train model
- Predict charges
- Compare with actual values
- Display results

Results

- The machine learning model successfully predicted courier charges with high accuracy
- Random Forest model performed best compared to other models
- Achieved high performance with R^2 score around 0.90+
- Identified key factors such as weight, distance, and service type affecting charges
- Detected deviations between estimated and actual charges
- Reduced billing errors and discrepancies
- Visualization (graphs/dashboards) provided clear insights into pricing patterns

Research Paper (Submitted Proof)

Submissions

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Paper ID	Title	Files	Status	Actions
1142	Preliminary Analysis of B2B Courier Charge Accuracy Using ML Show abstract	Submission files: ④ B2B_Courier_Charge_ML_IEEE_modified (1).pdf	Awaiting Decision	Submission: Edit Submission Edit Conflicts Delete Submission

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Paper ID	Title	Files	Status	Actions
1142	Preliminary Analysis of B2B Courier Charge Accuracy Using ML Show abstract Accurate estimation and validation of courier charges are essential for B2B logistics operations as they help maintain cost control, ensure billing transparency, and build trust between logistics providers and business clients. In large supply chain systems, courier pricing is influenced by multiple factors such as shipment weight, transportation distance, service type, delivery schedule, and carrier-specific pricing rules. Differences between estimated and actual billed charges can lead to revenue loss and operational inefficiencies. This research paper presents a preliminary study on improving the accuracy of B2B courier charge estimation using machine learning techniques. By analyzing historical shipment data, the study identifies key factors responsible for charge deviations and develops a predictive regression model to estimate final courier charges. A Random Forest Regressor is used to capture nonlinear relationships between shipment attributes and pricing, thereby improving prediction accuracy. The model's predicted charges are compared with actual billing records to identify discrepancies, and its performance is evaluated using standard regression metrics. Additionally, statistical summaries and graphical visualizations are used to interpret the results and provide meaningful insights for decision-making. The findings demonstrate that machine learning can support automated systems for charge validation and cost optimization in logistics operations. Overall, the study provides a foundation for building scalable, data-driven solutions that reduce billing discrepancies and enhance pricing transparency in B2B logistics systems.	Submission files: ④ B2B_Courier_Charge_ML_IEEE_modified (1).pdf	Awaiting Decision	Submission: Edit Submission Edit Conflicts Delete Submission

Conclusion

- Successfully applied Machine Learning for courier charge prediction.
- Used historical data to identify key factors:
 - Weight
 - Distance
 - Service type
 - Carrier details
- Random Forest model provided high accuracy
- Handles complex and non-linear relationships effectively
- Detects discrepancies between predicted and actual charges

Thank You